

# Scaling Laws for Softmax-Kernel In-Context Bayesian Near-Field Linear Sampling

Janis and Hancya

5 August 2026

## Abstract

We study how an in-context posterior-moment solver for the original two-dimensional near-field linear sampling method scales with the number  $n$  of training inverse problems, network size  $P$ , context length  $m$ , recurrent depth, and wall-clock time. The input is a complex multistatic response from deterministic point sources around one to six sound-soft obstacles. A fixed von Mises/softmax covariance selects the Gaussian-process feature space; its nonlinearity is a modelling choice and is not learned. Richardson, heavy-ball (HB), Chebyshev, population-PCG and context-PCG share the same Bayesian moment decoder and solve posterior mean and covariance right-hand sides in parallel. Context-PCG adds a prompt-conditioned SPD factor that is frozen during each solve. A hybrid starts from analytic angular-Jacobi and learns only a bounded residual log-gain. Identity-preconditioned CG is the matched-matrix-vector-product control. The sweep is supplemented by training-free physical Jacobi, angular block-Jacobi and GP-basis angular-Jacobi PCG controls. It contains 90 trained models, 810 dataset checkpoints and 362,880 learned-model/task evaluations. We report task-level uncertainty, inference time, empirical scaling exponents and a simultaneous distribution-free held-out risk bound. At  $m = 48$  and  $T = 32$ , context-PCG improves the common-coordinate physical residual by  $2.05\times$ , reduces the relative LSM-score error by  $6.44\times$ , and raises numerical 95% posterior-band coverage from 0.243 to 0.873. Across 768 independent acquisition batches its paired geometric-mean residual gain is  $2.24\times$  and it wins 96.4% of pairs, although analytic angular-Jacobi PCG remains faster on this small dense problem. The paired AP increase is only 0.0021, separating numerical fidelity from physical localization resolution. The main question is not whether a large controller can imitate CG, but when learned conditioning improves CG under severe, task-dependent ill-conditioning.

## 1 Problem and posterior moments

Let  $N_D \in \mathbb{C}^{m \times m}$  be the whitened near-field matrix for a sound-soft obstacle  $D$ ,  $\Phi = [\phi_z]_z$  the Green-function probes on a two-dimensional sampling grid, and  $K_\Gamma \succ 0$  a prescribed angular GP covariance. The nonlinear feature choice is

$$W_{ij}^\rho = \exp\{\rho(\cos(\theta_i - \theta_j))\}, \quad S_{ij}^\rho = \frac{W_{ij}^\rho}{\sum_\ell W_{i\ell}^\rho} = \text{softmax}_j[\rho(\cos(\theta_i - \theta_j))],$$

$$K_\Gamma = (1 - \eta)I + \eta D_W^{-1/2} W^\rho D_W^{-1/2}, \quad D_W = \text{diag}(W^\rho \mathbf{1}).$$

The experiments fix  $\rho(t) = t$  and  $\eta = 0.2$ . The function  $\rho$  is a kernel-specific modelling choice that selects the GP feature space; it is not an attention temperature fitted by the network. Admissible choices must make  $W^\rho \succeq 0$ ; for the chosen von Mises kernel the symmetric normalization in  $K_\Gamma$  preserves a Hermitian positive GP covariance. On a uniform full aperture its constant row sum makes it equivalent to the symmetric form of the row-softmax weights. Define

$$H_D = N_D K_\Gamma N_D^* + I, \quad H_D Q_\mu = \Phi, \quad H_D Q_\Sigma = N_D K_\Gamma.$$

One tied recurrence acts on  $[\Phi, N_D K_\Gamma]$  and yields

$$M_D = K_\Gamma N_D^* Q_\mu, \quad \Sigma_D = K_\Gamma - K_\Gamma N_D^* Q_\Sigma.$$

Thus a small mean residual is insufficient for Bayesian correctness: the covariance residual must converge as well. This shares the posterior-distribution ICL viewpoint of Kang, Lee and Cheng [2], but the learned object here is an iterative conditioner for a prescribed nonlinear GP feature space. We also report the fraction of grid scores for which the direct-solve reference lies within the approximate  $\mu \pm 1.96\sigma$  band. This is a numerical-UQ fidelity diagnostic: it checks whether iterative error is small relative to the reported posterior width, but is not frequentist coverage of the

obstacle. This is the original active near-field LSM with all deterministic point sources fired at every acquisition. It does not replace  $N_D$  by the cross-correlation, volume or imaginary near-field operators used for random sources, nor by the asymptotic small-random-scatterer data model [3, 4]. Randomness below concerns obstacles, additive noise and held-out acquisition geometry, not the physical source model.

## 2 Variable-context architecture and matched solvers

The covariance block uses a deterministic orthonormal harmonic sketch with a fixed number of columns and  $m$  rows. Its feature dimension is therefore independent of context length, while the physical operator is never padded or replaced by a learned surrogate. Training contexts are [12, 16, 24, 32]; evaluation contexts are [8, 12, 16, 24, 32, 48], so the endpoints include context-length extrapolation. Controller widths are [32, 64, 128, 256, 512] and sample sizes are [128, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768].

At test depth  $T = 32$ , all recurrent methods use exactly  $T$  applications of the transformed operator for both posterior blocks. Identity-CG sets the population factor to  $I$ . Population-PCG learns a positive geometry/population factor and retains exact task-wise Krylov coefficients. Context-PCG augments that factor with a permutation-equivariant token encoder of the current near-field Hessian. It predicts positive modal gains in the fixed angular-kernel basis once per prompt; the factor is then held fixed for all  $T$  iterations, so this is standard PCG rather than flexible CG. Writing the fixed receiver angular-feature kernel as  $G_\Gamma = U_\Gamma \Lambda_\Gamma U_\Gamma^*$ , its factor has the form

$$C_\theta(D, \Gamma) = U_\Gamma \text{diag}(g_\theta(U_\Gamma^* H_D U_\Gamma))^{1/2} U_\Gamma^*, \quad g_\theta > 0.$$

The task token for each angular mode contains diagonal, row- $\ell_1$  and row- $\ell_2$  statistics of  $U_\Gamma^* H_D U_\Gamma$ ; shared token maps and mean pooling make the controller valid at variable  $m$ . The analytic-learned hybrid instead uses

$$g_{\text{hyb},j} = \frac{\exp\{m^{-1} \sum_k \log(U_\Gamma^* H_D U_\Gamma)_{kk}\}}{(U_\Gamma^* H_D U_\Gamma)_{jj}} \exp\{\delta_{\theta,j}\}, \quad |\delta_{\theta,j}| \leq \frac{1}{2}.$$

The zero-initialized residual network therefore begins exactly at the training-free angular-Jacobi factor and can refine it only inside a prescribed multiplicative trust region; its regularizer acts on  $\delta_\theta$ , not on the analytic base. Looped HB [5], Chebyshev and Richardson additionally infer finite-horizon spectral endpoints from RHS-weighted Krylov statistics. Consequently, PCG versus CG isolates learned conditioning; a looped method versus PCG measures whether a stationary learned recurrence is worth replacing Krylov adaptation. Post-training controls construct Jacobi factors directly from the current  $H_D$ : its physical diagonal, contiguous angular blocks, or the diagonal of  $U_\Gamma^* H_D U_\Gamma$ . The last is the non-learned operator-aware baseline for the same fixed GP feature basis used by context-PCG. With  $G$  spatial probes, one recurrent step costs  $O(m^2(G+m))$  for the concatenated posterior blocks and stores  $O(m(G+m))$  state; depth  $T$  multiplies the first term. A dense direct solve adds an  $O(m^3)$  factorization. The empirical timing curves retain forward construction and decoding overhead, so their slope need not equal the pure matrix-matrix asymptote at small  $m$ . For any fixed SPD factor  $C_D$ , standard PCG gives [6]

$$\frac{\|e_T\|_{H_D}}{\|e_0\|_{H_D}} \leq 2 \left( \frac{\sqrt{\kappa_D} - 1}{\sqrt{\kappa_D} + 1} \right)^T, \quad \kappa_D = \kappa(C_D^* H_D C_D).$$

The condition-number audit below therefore supplies the missing link from an in-context factor to a depth-dependent solver certificate. We distinguish two residual norms. For  $q_T = C_D x_T$  and right-hand side  $b$ ,

$$r_{\text{tr}} = \frac{\|C_D^*(b - H_D q_T)\|_F}{\|C_D^* b\|_F}, \quad r_{\text{phys}} = \frac{\|b - H_D q_T\|_F}{\|b\|_F}.$$

The online objective and original dataset/width/context scaling grid store  $r_{\text{tr}}$ , because that is the residual seen by the transformed solver. Since its norm changes with  $C_D$ , every post-training CG comparison, depth curve, tolerance frontier and geometry/stress audit below instead uses  $r_{\text{phys}}$ . Score error and Bayesian coverage are coordinate independent. Both residuals are retained in the released CSV files. To identify the intended factor rather than merely a visually acceptable LSM ranking, context-PCG also minimizes

$$\mathcal{L}_{\text{id}}(D) = \frac{1}{m} \left\| I - \tilde{H}_{\theta,D} \right\|_F^2, \quad \tilde{H}_{\theta,D} = \frac{C_\theta^* H_D C_\theta}{\|C_\theta^* H_D C_\theta\|_\infty}.$$

Indeed,  $\varepsilon_D = \|I - \tilde{H}_{\theta,D}\|_2 \leq \sqrt{m \mathcal{L}_{\text{id}}(D)}$ . Whenever  $\varepsilon_D < 1$ , all eigenvalues lie in  $[1 - \varepsilon_D, 1 + \varepsilon_D]$  and  $\kappa_D \leq (1 + \varepsilon_D)/(1 - \varepsilon_D)$ . This is the explicit identification-loss  $\Rightarrow$  spectral certificate  $\Rightarrow$  PCG-depth chain; ranking loss alone has no such implication.

Table 1: Held-out physical scenarios. Noise is relative complex measurement noise; aperture is in degrees.

| Scenario             | obstacles | shape | $k$  | noise | aperture |
|----------------------|-----------|-------|------|-------|----------|
| ID one obstacle      | 1         | mixed | 8.0  | 0.10  | 360      |
| ID four obstacles    | 4         | mixed | 8.0  | 0.15  | 360      |
| OOD six obstacles    | 6         | mixed | 8.0  | 0.15  | 360      |
| OOD stars            | 3         | star  | 8.0  | 0.15  | 360      |
| OOD 30 percent noise | 4         | mixed | 8.0  | 0.30  | 360      |
| OOD half aperture    | 3         | mixed | 8.0  | 0.15  | 180      |
| OOD wavenumber 12    | 3         | mixed | 12.0 | 0.15  | 360      |

Table 2: Empirical exponents in Eq. (1) for both risks; positive data/width exponents indicate improvement. The context exponent is signed:  $\beta < 0$  records fixed-depth degradation as the linear system grows.

| Method            | risk         | data $\alpha$ | context $\beta$ | width $\gamma$ | $R^2$  |
|-------------------|--------------|---------------|-----------------|----------------|--------|
| context-PCG       | localization | 0.020         | 3.000           | 0.010          | 0.983  |
| context-PCG       | solver       | 1.264         | -2.361          | 3.000          | 0.545  |
| hybrid-PCG        | localization | 0.010         | 3.000           | 0.010          | 0.983  |
| hybrid-PCG        | solver       | 1.816         | -2.362          | 3.000          | 0.981  |
| looped-Chebyshev  | localization | 1.130         | 3.000           | 0.010          | 0.247  |
| looped-Chebyshev  | solver       | 0.424         | -1.156          | 0.628          | 0.721  |
| looped-HB         | localization | 0.876         | 3.000           | 0.010          | 0.087  |
| looped-HB         | solver       | 0.345         | -0.997          | 0.726          | 0.850  |
| looped-Richardson | localization | 0.371         | -0.138          | 0.010          | -0.242 |
| looped-Richardson | solver       | 0.195         | -0.915          | 0.590          | 0.718  |
| population-PCG    | localization | 0.010         | 3.000           | 0.011          | 0.984  |
| population-PCG    | solver       | 1.586         | -2.443          | 3.000          | 0.775  |

### 3 Axes and scaling model

We adopt the depth–width–context–time resource axes of Bordelon, Letey and Pehlevan [1]. Their exact proportional-limit asymptotics apply to deep *linear* self-attention regression with specified covariance ensembles. The nonlinear fixed-softmax GP feature map, complex obstacle-dependent near-field operator and Bayesian covariance solve here violate those assumptions; importing their closed risk equations would therefore be unjustified. We retain the resource decomposition and test an explicitly descriptive finite-grid law instead. The dataset size  $n$  is the number of unique physical inverse problems presented once in an online stream; the eight examples in an optimizer batch share one sampled acquisition configuration. The context length  $m$  is both the number of point sources and receivers, giving  $m^2$  complex measurements. Network size  $P$  counts trainable parameters. Each held-out configuration contains fixed, never-trained MFS tasks spanning ID obstacles and OOD obstacle count, shape, noise, aperture and frequency.

For localization risk  $R_{\text{loc}} = 1 - \text{AP}$  and bounded numerical risk  $R_{\text{solve}} = r_{\text{tr}}/(1 + r_{\text{tr}})$ , we separately fit the descriptive law

$$R(n, m, P) = R_{\infty} + A(n/n_0)^{-\alpha} + B(m/m_0)^{-\beta} + C(P/P_0)^{-\gamma} + D\sqrt{(P/P_0)/(n/n_0)}. \quad (1)$$

The last term allows a finite-data width penalty. We constrain  $\alpha, \gamma > 0$  but let  $\beta$  be signed: more measurements can improve statistical localization while simultaneously making a fixed-depth solve harder through condition growth. Equation (1) is an empirical scaling model, not a theorem.

### 4 A simultaneous held-out generalization bound

Let  $L_c \in [0, 1]$  be localization risk for a pre-specified trained configuration  $c = (n, m, P, \text{method})$ , and let  $\hat{R}_c$  average  $T_c$  independent held-out tasks. A union bound and one-sided Hoeffding inequality give the following non-asymptotic certificate.

**Theorem 1** (Simultaneous task-distribution bound). *For  $K$  evaluated configurations, with probability at least  $1 - \delta$ , simultaneously for every  $c$ ,*

$$R_c \leq \hat{R}_c + \sqrt{\frac{\log(K/\delta)}{2T_c}}.$$

We use  $\delta = 0.05$  and report the clipped upper bound in Figure 1. Training samples determine the learned hypothesis and hence  $\widehat{R}_c$ ; held-out sample size controls certification slack. Because seed is pooled rather than included in  $c$ , the certificate concerns the mean risk of the three evaluated training seeds, not a guarantee for each seed separately. The same theorem is applied to the bounded numerical risk  $L_c^{\text{solve}} = r_c^{\text{tr}} / (1 + r_c^{\text{tr}})$ , where  $r_c^{\text{tr}}$  is the transformed-coordinate posterior-mean residual stored by the training grid. This second risk remains sensitive after localization AP saturates. This distinction matters. Replacing  $T_c$  by training size  $n$  after fitting would be invalid, while a generic parameter-count uniform bound for this nonconvex complex network is vacuous at the present  $P/n$  ratios.

We additionally draw independent acquisition batches rather than relabel tasks from one geometry. Each bounded observation averages four fresh obstacle and noise realizations at one newly rotated/jittered deterministic point-source array. Every per-scenario configuration has at least 192 independent geometry batches over the three frozen training seeds. Applying the same union-bound argument to these batch means yields Figure 2; its largest slack is 0.140. This remains a simulator-distribution certificate, not a claim about the random-source LSM. At  $m = 48$ , context-PCG reduces the geometry-batch mean residual by  $2.040\times$  relative to CG, with a worst scenario-wise gain of  $1.277\times$ . Its numerical 95% band coverage is 0.914, versus 0.186 for CG. The paired AP difference is only  $2.06\text{e-}03$  (95% interval  $[1.45\text{e-}03, 2.66\text{e-}03]$ ), whereas the paired numerical-coverage increase is 0.728 ( $[0.718, 0.737]$ ). Thus the large numerical gain mainly restores solver and UQ fidelity rather than changing the physical localization ranking. All 12 seed/scenario cells retain the advantage, with minimum gain  $1.149\times$ . On the same independent geometries, angular-Jacobi, hybrid-PCG and context-PCG have mean residuals 0.559, 0.566, and 0.501, respectively. The angular-to-context ratio is 1.116, with worst scenario ratio 0.983. Their angular and hybrid numerical coverages are 0.874 and 0.867. Pairing methods on each independent geometry batch gives an angular-to-context geometric-mean gain of 1.103, with 95% normal-approximation log-ratio interval  $[1.089, 1.117]$ . Context-PCG wins 0.701 of paired batches (Wilson 95% interval  $[0.667, 0.732]$ ).

For a comparable curve, every dataset checkpoint in Figure 1 uses exactly 8 tasks per scenario and seed. The largest checkpoint is additionally audited with 32 tasks per scenario; those tighter values are stored separately and used in the final performance table.

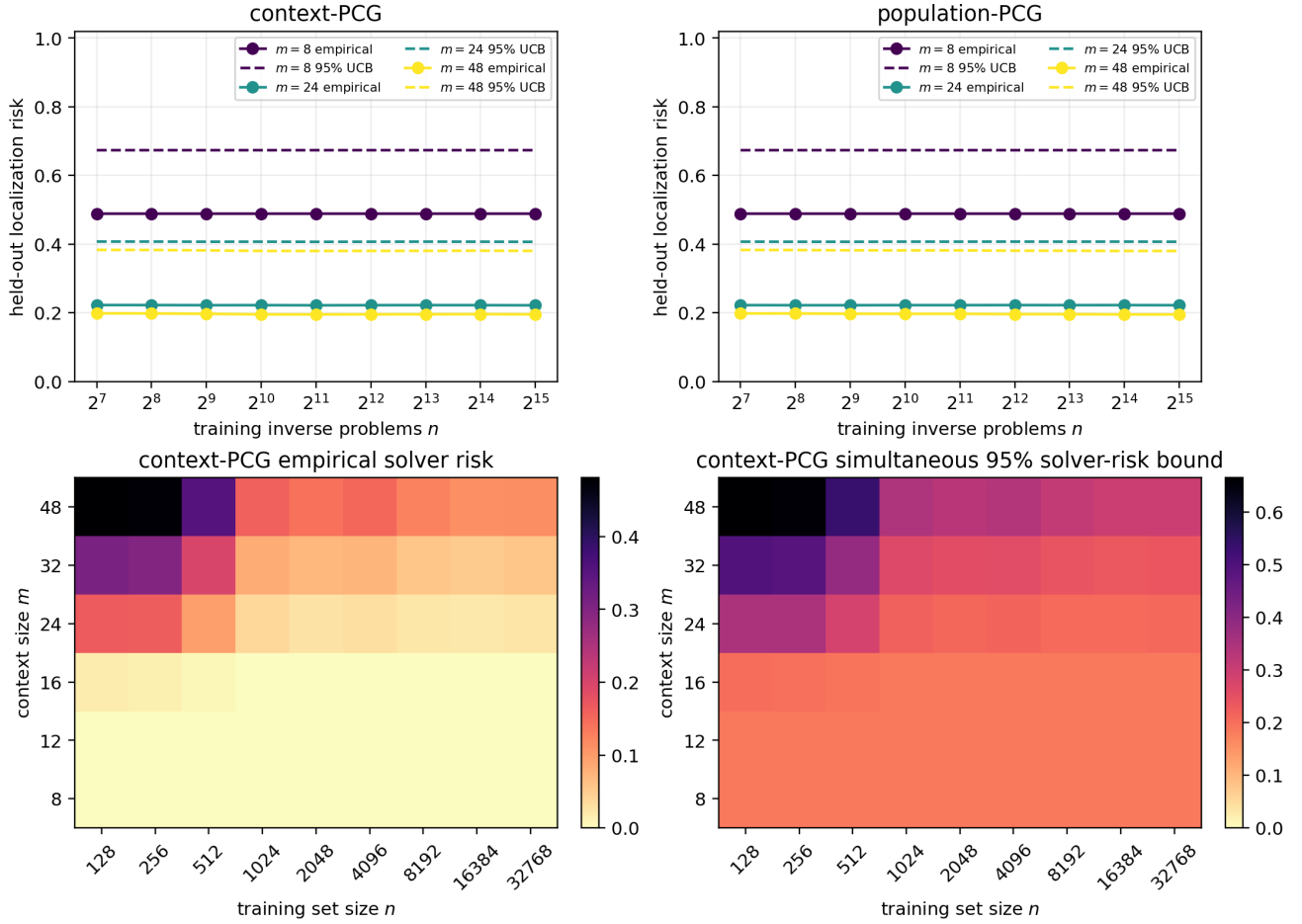


Figure 1: Empirical localization and numerical solver risks with simultaneous 95% held-out upper bounds across dataset and context sizes.

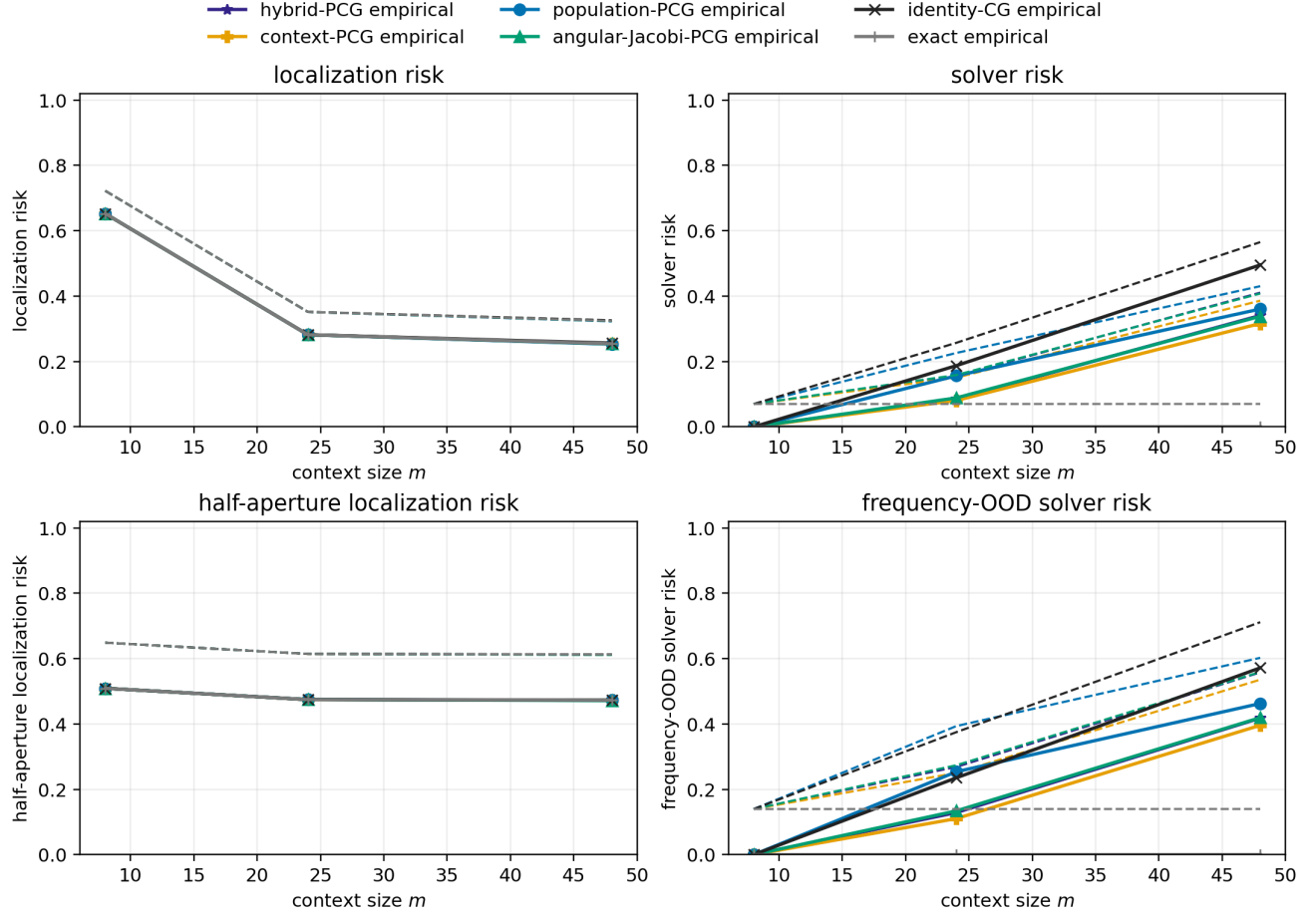


Figure 2: Independent-acquisition generalization. Each bounded observation is the mean of four fresh obstacle/noise tasks sharing one independently rotated and jittered deterministic source-receiver geometry. Dashed curves are simultaneous 95% Hoeffding upper bounds.

## 5 Results

On the original scaling grid, prompt-conditioned PCG lowers the transformed-coordinate posterior-mean residual relative to CG in 7 of 7 held-out scenarios at  $m = 48$ . Its worst scenario-wise gain is  $4.126\times$  and its median gain is  $7.534\times$ . All 21 individual training-seed/scenario cells favor context-PCG; the smallest such gain is  $3.507\times$ . The largest absolute AP change is only  $8.20e-03$ ; hence the learned conditioner is substantially more accurate in score fidelity while localization effectiveness remains essentially tied at the physical point-spread limit. At  $m = 48$ , context-PCG attains numerical 95% band coverage 0.865, compared with 0.244 for CG. This is UQ fidelity to the exact Bayesian solve, not a physical coverage claim. The foundation-model width sweep also rejects a “bigger is automatically better” interpretation. Width 64 gives the smallest extrapolative transformed residual, while width 512 has  $22.451\times$  as many parameters and a  $1.035\times$  larger residual. The controller has reached a feature/basis plateau rather than a parameter-limited regime.

Across the 20 stress levels, the context-PCG gain over CG ranges from  $1.071\times$  to  $4.321\times$ ; its smallest paired-batch win rate at any level is 0.625. The first sixteen levels vary one factor; the four joint-severity levels co-vary obstacle count, noise, aperture, frequency and acquisition jitter. These remain simulator perturbations, not a certificate for arbitrary distribution shift. Context-PCG has the smallest residual in 11 of the 20 levels, versus 1 for angular-Jacobi, 3 for population-PCG and 2 for the hybrid, while looped HB wins 3. Robustly beating CG therefore does not imply dominance over every informed preconditioner. Across all stress batches, context-PCG’s geometric-mean residual gain is  $2.172\times$ , with AP and numerical-coverage changes  $1.08e-03$  and 0.539 relative to CG. Looped HB attains a residual gain of  $0.723\times$  but changes AP by  $-0.300$  and coverage by  $-0.257$ . Its occasional short-horizon residual win is therefore not a reconstruction or UQ win. At the most extreme joint shift, context-PCG retains a residual gain of  $1.958\times$  (95% interval  $[1.870, 2.050]$ ) and wins 0.974 of paired batches (Wilson 95% interval  $[0.940, 0.989]$ ). Angular-Jacobi retains  $1.337\times$ , whereas population-PCG and looped HB fall to  $0.418\times$  and  $0.120\times$ . The context-PCG coverage gain is nevertheless 0: this is residual robustness, not universal UQ calibration.

At the extreme joint shift, the raw median condition number is  $3.61e+03$ . Context-PCG reduces it by  $5.448\times$  to 589.921, whereas angular-Jacobi reduces it by  $5.387\times$  to 588.200. The commutator is 0.329. Context and angular-Jacobi therefore have nearly identical scalar condition numbers but residual gains  $1.958\times$  and  $1.337\times$ . Condition number alone does not rank the finite-horizon solve; spectral clustering and right-hand-side alignment matter, while the nonzero commutator still limits both fixed-basis factors.

The equal-budget transformed-coordinate residual ratios at the final central configuration are  $3.407\times$  for the posterior mean and  $0.985\times$  for posterior covariance. Ratios above one mean that the learned factor lowers the transformed-coordinate optimization metric. Since the coordinate factor differs by method, these ratios are not interpreted as physical solver gains; the common-coordinate audit below supplies that comparison. Localization can nevertheless saturate before residual convergence; in that regime additional solver accuracy does not overcome the physical point-spread limit. Across all six contexts, the corresponding gains are  $4.336\times$  and  $1.535\times$ . At the extrapolative context  $m = 48$  they reach  $4.703\times$  and  $1.909\times$ , respectively. Thus the transformed training-metric separation grows with context, although covariance separation is not uniform. Whether this survives in the physical coordinate is evaluated separately below.

At the same central configuration, prompt-conditioned PCG improves ordinary CG in transformed-coordinate residual by  $8.030\times$  on the posterior mean and  $1.785\times$  on posterior covariance. Across all contexts its corresponding gains are  $7.936\times$  and  $2.183\times$ . This is a diagnostic of whether the current operator carries useful in-context signal within the prescribed optimization coordinates; it is not the direct physical CG claim. The post-training  $r_{\text{phys}}$  audit below provides the coordinate-invariant test. Within the shared model implementation, the price is controller overhead: context-PCG costs  $1.144\times$  the CG batch time at  $m = 24$  and  $1.269\times$  at  $m = 48$ . It is therefore iteration-efficient, not a large wall-clock speedup at fixed depth. After removing the unused endpoint controller from PCG inference, the iterative methods are faster than dense factorization on the tested batches. The optimized classical controls below provide the stricter timing comparison.

We also test an analytic-learned hybrid: the current Hessian is rotated into the prescribed GP angular feature basis, its diagonal supplies a training-free positive preconditioner, and a zero-initialized context network learns only a multiplicative residual correction. At the central configuration it improves CG in transformed-coordinate residual by  $7.500\times$  for the mean and  $1.751\times$  for covariance. Across all contexts the corresponding gains are  $6.116\times$  and  $2.041\times$ . At  $m = 48$  and  $T = 32$ , however, its physical mean residual is 0.630, whereas context-PCG attains 0.521 and analytic angular-Jacobi attains 0.629. The angular-to-hybrid residual ratio is 0.997; at the common central width this tests whether the correction improves the analytic base without extra capacity. Across the width sweep, the best hybrid uses width 64 and attains transformed residual 0.178, covariance residual 0.014, and coverage 0.837. These width-sweep residuals are in transformed coordinates and are therefore not divided by the physical residual of the analytic control; their homogeneous physical comparison is the matched-width, matched-depth audit above. This

Table 3: Final comparison at context  $m = 24$ , width 128 and  $n = 32768$ . Baselines do not train. Runtime is milliseconds per held-out batch.

| Method            | AP    | $r_{tr}$ mean | $r_{tr}$ cov. | score error | 95% cov. | ms     |
|-------------------|-------|---------------|---------------|-------------|----------|--------|
| hybrid-PCG        | 0.774 | 0.030         | 4.16e-03      | 1.03e-03    | 0.970    | 8.372  |
| context-PCG       | 0.774 | 0.028         | 4.09e-03      | 1.01e-03    | 0.968    | 8.338  |
| population-PCG    | 0.774 | 0.065         | 7.41e-03      | 2.05e-03    | 0.943    | 7.548  |
| looped-HB         | 0.505 | 0.127         | 0.020         | 0.104       | 0.167    | 20.629 |
| looped-Chebyshev  | 0.651 | 0.093         | 0.028         | 0.125       | 0.225    | 17.850 |
| looped-Richardson | 0.300 | 0.259         | 0.074         | 0.256       | 0.045    | 17.597 |
| identity-CG       | 0.774 | 0.222         | 7.29e-03      | 4.44e-03    | 0.955    | 7.289  |
| exact             | 0.774 | 6.21e-05      | 1.67e-06      | 0           | 1.000    | 10.909 |

Table 4: Scenario-wise comparison at the extrapolative context  $m = 48$ , width 128 and  $n = 32768$ . Gains use the transformed-coordinate residual  $r_{tr}$ ; values above one favor context-PCG.

| Scenario             | AP CG | AP C-PCG | $r_{tr}$ CG | $r_{tr}$ C-PCG | mean gain | cov. gain |
|----------------------|-------|----------|-------------|----------------|-----------|-----------|
| ID four obstacles    | 0.809 | 0.812    | 0.963       | 0.129          | 7.460     | 2.939     |
| ID one obstacle      | 0.987 | 0.988    | 2.070       | 0.140          | 14.820    | 1.489     |
| OOD 30 percent noise | 0.781 | 0.782    | 0.415       | 0.045          | 9.126     | 4.776     |
| OOD half aperture    | 0.546 | 0.545    | 0.966       | 0.234          | 4.126     | 1.280     |
| OOD six obstacles    | 0.775 | 0.776    | 0.843       | 0.103          | 8.180     | 4.402     |
| OOD stars            | 0.812 | 0.821    | 0.859       | 0.114          | 7.534     | 2.366     |
| OOD wavenumber 12    | 0.856 | 0.861    | 1.311       | 0.194          | 6.773     | 2.362     |

control is the criterion for deciding whether preconditioning needs to be learned rather than merely computed in context.

The post-training depth audit contains 71,568 task-level evaluations over  $T \in [1, 2, 4, 8, 16, 32, 48, 64, 96, 128]$ . At the extrapolative context  $m = 48$ , the CG-to-population-PCG physical posterior-mean residual gain is  $1.585\times$  at  $T = 32$  and  $2.119\times$  at  $T = 64$ ; the corresponding prompt-conditioned gains are  $2.049\times$  and  $2.641\times$ . At the same two depths, context-PCG improves looped HB by  $1.887\times$  and  $9.771\times$ , respectively; HB has a competitive short-horizon transient but its stationary recurrence does not retain the long-depth Krylov convergence. At  $m = 48$  and  $T = 32$ , looped HB has mean residual 0.983, average precision 0.478, and numerical 95% coverage 0.095; its batch runtime is 83.616 ms. Its relative LSM-score error is 0.139, versus 0.029 for CG and  $4.54e-03$  for context-PCG. In an ill-conditioned system, a smaller Euclidean residual can still leave larger solution error along small-eigenvalue directions; this explains why HB’s residual transient does not produce a good reconstruction. Thus its residual improvement over CG does not translate into faithful LSM localization or calibrated Bayesian uncertainty, and its dense learned update is substantially more expensive than Krylov PCG. At  $T = 128$ , the equal-depth CG-to-context-PCG gain reaches  $13.847\times$ ; this ratio is taken near the numerical convergence floor and is not a claim of improved physical localization. For a tolerance-oriented comparison at  $m = 48$ , context-PCG first reaches mean residual 0.1 at  $T = 64$ , versus  $T = 96$  for CG; for residual 0.01 the depths are 96 and 128. At the 0.01 target, however, the measured batch times are 23.128 ms and 24.912 ms, respectively: the optimized implementation preserves a modest wall-clock benefit from the iteration-count saving at this matrix size. Figure 12 therefore distinguishes a genuine conditioning advantage from merely granting the learned method more iterations. The fact that depth remains useful at large  $m$  is qualitatively consistent with the randomly rotated structured-covariance regime of Bordelon et al.; it is not an assertion that their linear-model asymptotics govern this inverse-scattering system.

At  $m = 48$ , the median condition-number reduction is  $10.473\times$  for population-PCG and  $23.220\times$  for context-PCG. Figure 13 directly checks whether smaller residuals track spectral compression and whether the obstacle-dependent operator commutes with the fixed GP geometry. The analytic-learned hybrid reduces the condition number by  $20.328\times$  and leaves median condition number  $1.25e+03$ . The raw median condition number is still  $2.66e+04$ , and context-PCG leaves  $1.09e+03$ ; the normalized commutator 0.221 therefore identifies the remaining inability of fixed modal gains to rotate task-dependent eigendirections.

The training-free angular-Jacobi control is the critical test of whether the available in-context signal actually needs to be learned. At  $m = 48$  and  $T = 32$ , its physical posterior-mean residual is 0.629, versus 0.521 for context-PCG, a classical-to-learned ratio of 1.207. At  $T = 96$  the ratio is instead 0.243, so the analytical control overtakes the learned finite-horizon factor at high accuracy. Their numerical 95% band coverages at  $T = 32$  are 0.851 and 0.873, respectively. Its transformed condition number is  $1.01e+03$  and its runtime is 9.836 ms, or  $0.715\times$  the learned method. Removing unused feature/controller calculations from CG also reduces its  $T = 32$  runtime to 8.809 ms;

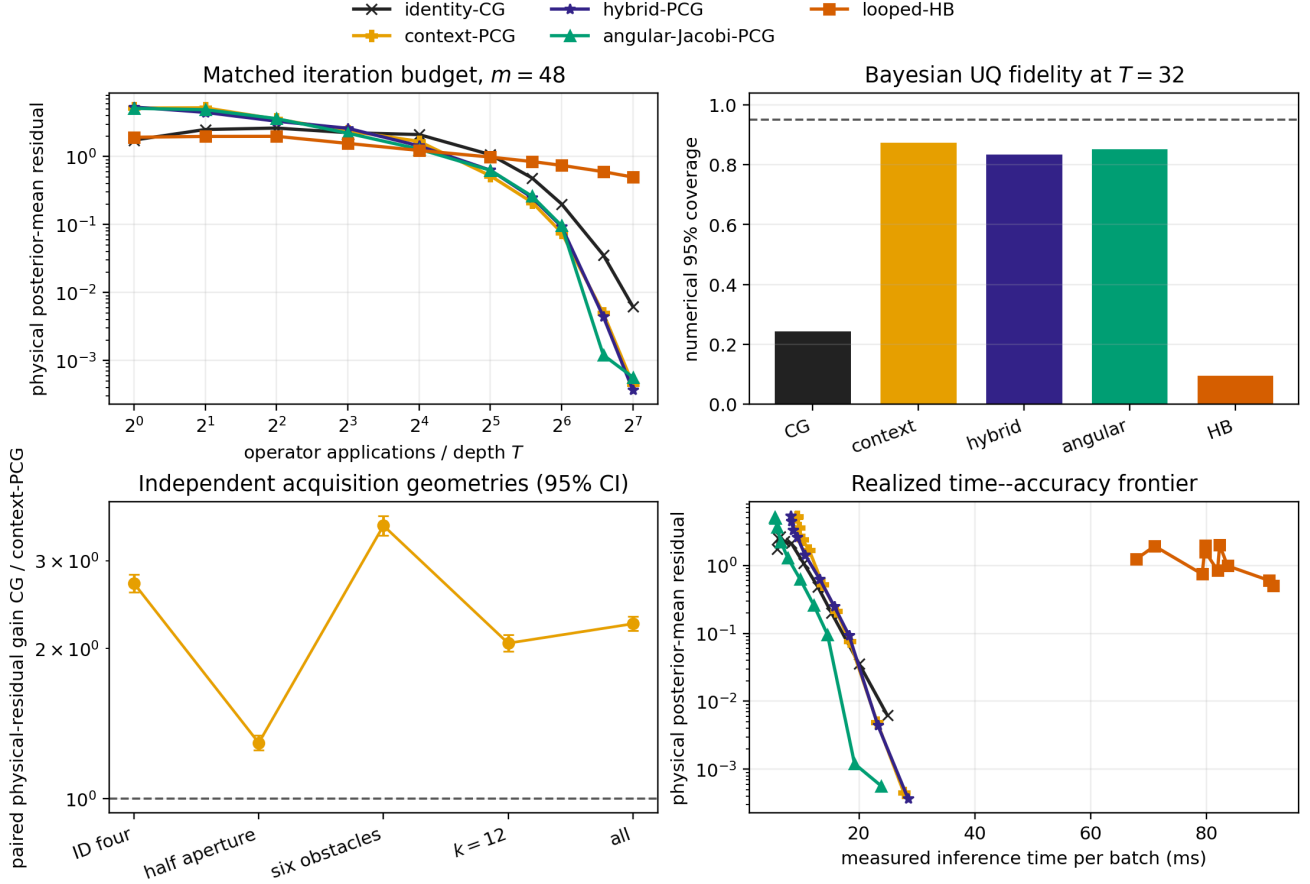


Figure 3: Direct comparison with ordinary CG. Top left: matched operator applications. Top right: numerical Bayesian coverage relative to the exact solve. Bottom left: paired physical-residual gains on independent acquisition geometries. Bottom right: measured time-accuracy paths. Values above one in the gain panel favor context-PCG.

context-PCG is then  $1.561\times$  slower. At residual target 0.01, angular-Jacobi reaches the target at  $T = 96$  in 19.240 ms; optimized CG uses  $T = 128$  and 22.862 ms, while context-PCG uses 23.128 ms. Across all 18 context/target cells in Figure 15, angular-Jacobi is fastest in 11, hybrid-PCG in 0, population-PCG in 0, identity-CG in 0, optimized CG in 5, block-Jacobi in 2 and context-PCG in 0. Hence context-PCG’s equal-depth advantage does not become a strict wall-clock win against every optimized operator-aware baseline. Thus the experiment supports learning the conditioner only to the extent that it improves or amortizes this cheap operator-aware baseline; diagonal and local block-Jacobi in the physical ordering do not extract the same angular GP structure.

The fitted powers must also be read diagnostically rather than as universal laws. For population-PCG the localization fit has  $R^2 = 0.984$ , whereas the solver-risk fit has  $R^2 = 0.775$ . A low or negative  $R^2$  rejects the proposed monotone separable power-law description on this finite grid; it is evidence of non-monotone optimization or a performance plateau, not a scaling exponent to extrapolate. Exponents reported as 3.000 hit the fit’s upper constraint and must be read as boundary estimates, not identified power laws.

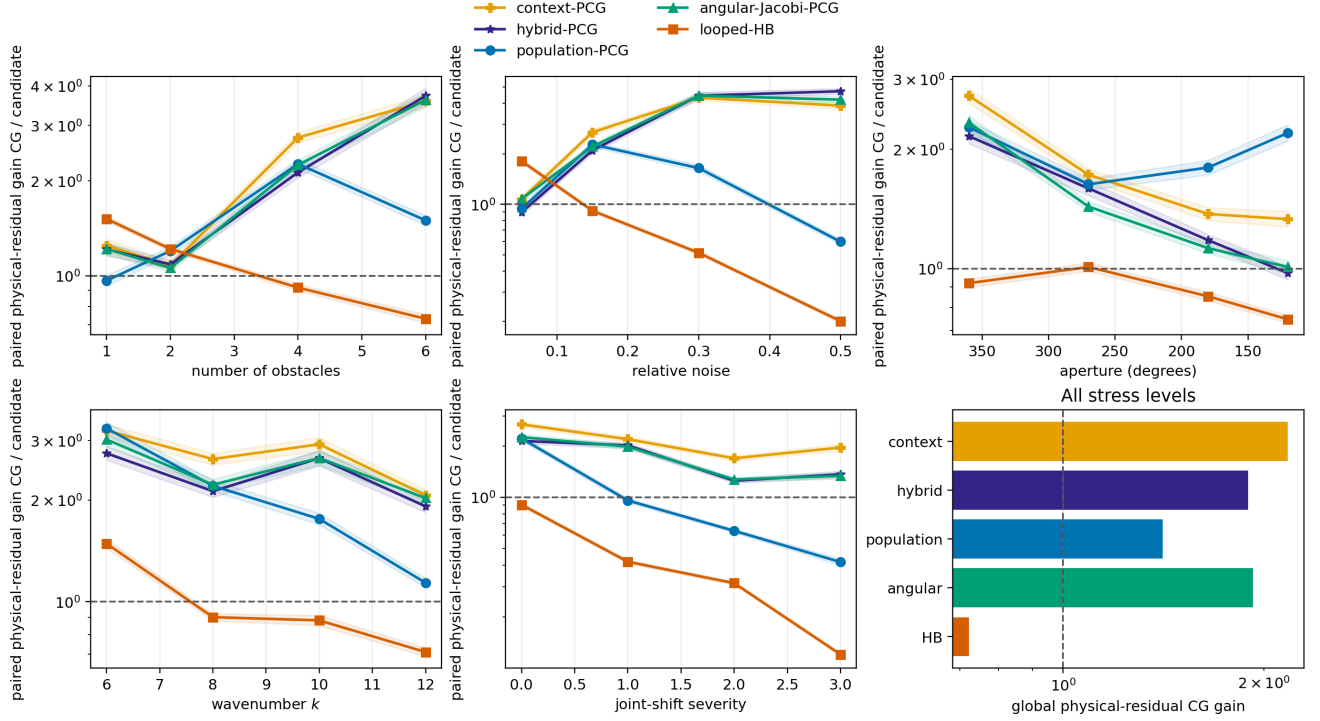


Figure 4: Marginal and joint physical stress sweeps at  $m = 48$  and  $T = 32$ . Curves are paired geometric-mean physical-coordinate residual gains over at least 192 independent acquisition batches per level; shaded regions are pointwise 95% normal-approximation log-ratio intervals. Values above one favor the preconditioned method over ordinary CG.

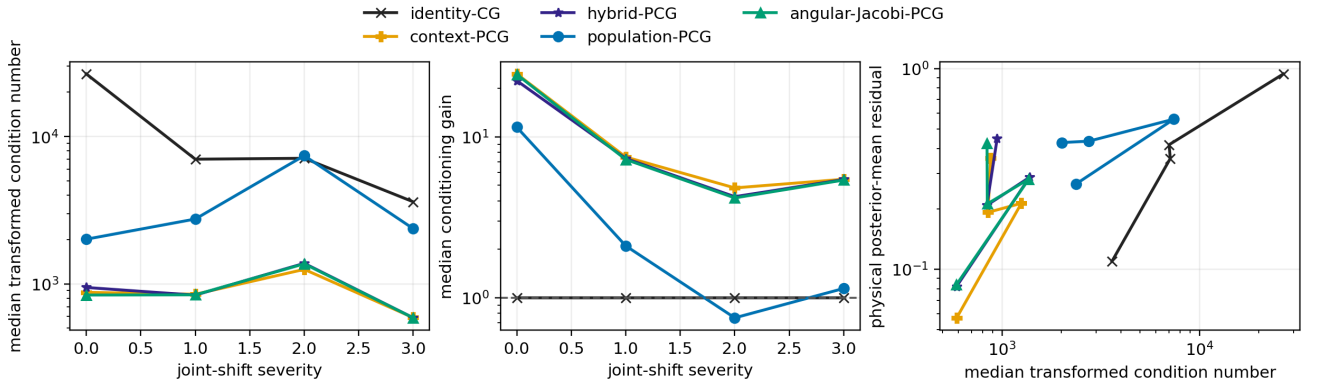


Figure 5: Spectral mechanism across the four joint-shift severities. Each point aggregates independent acquisition batches; the right panel links condition compression to the observed finite-depth physical residual.

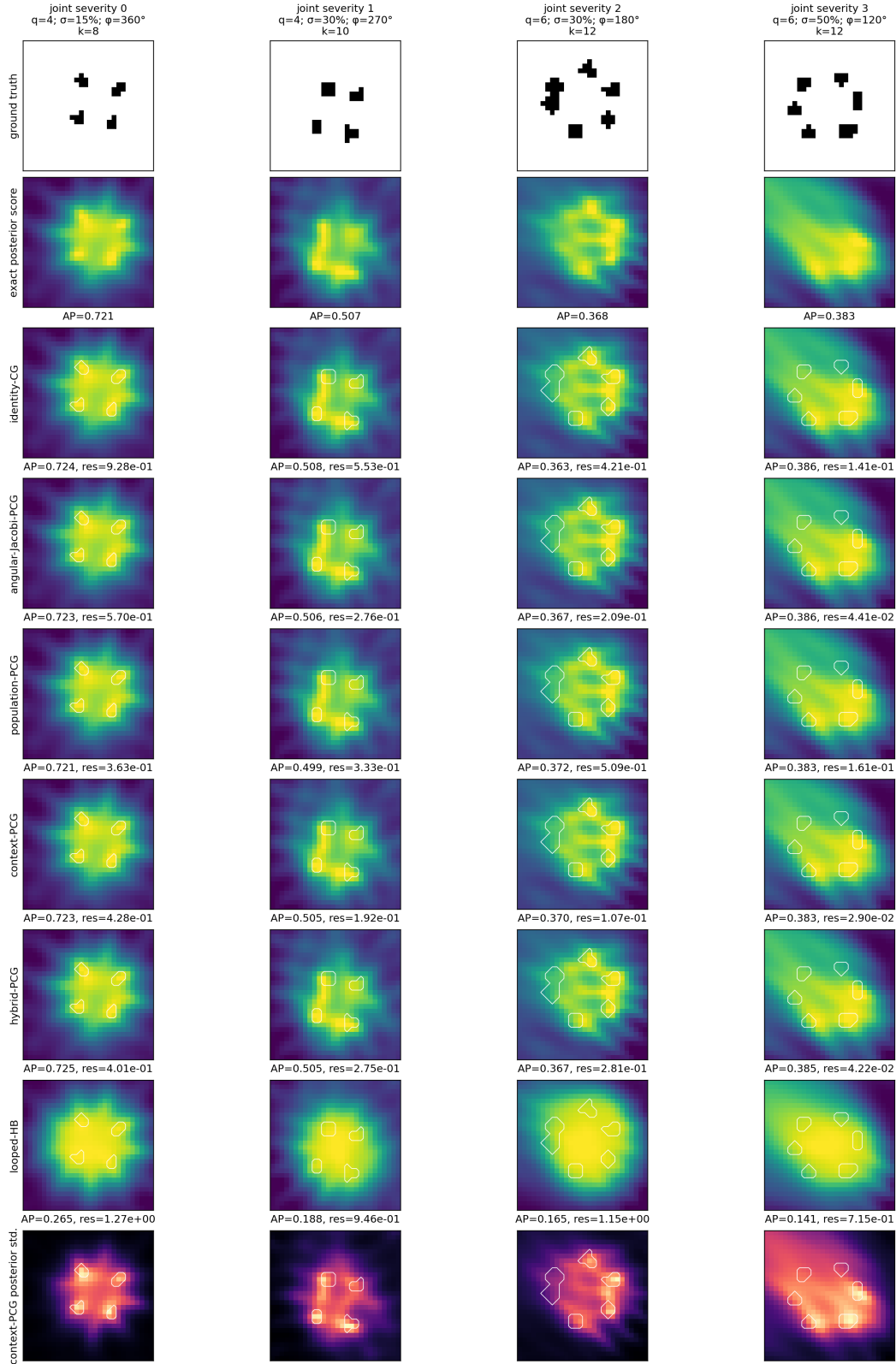


Figure 6: Pre-specified reconstructions along the joint-shift path (seed 17, geometry draw zero, task zero); no task was selected by performance. White contours show the true obstacles. Each score map is normalized by its own 1st–99th percentile range to compare localization shape; AP and physical residual are reported separately. Context-PCG preserves the exact LSM ranking while lowering the finite-depth residual, whereas looped HB has poor ranking despite some residual improvement. The last row is context-PCG posterior standard deviation, not a reconstruction score.

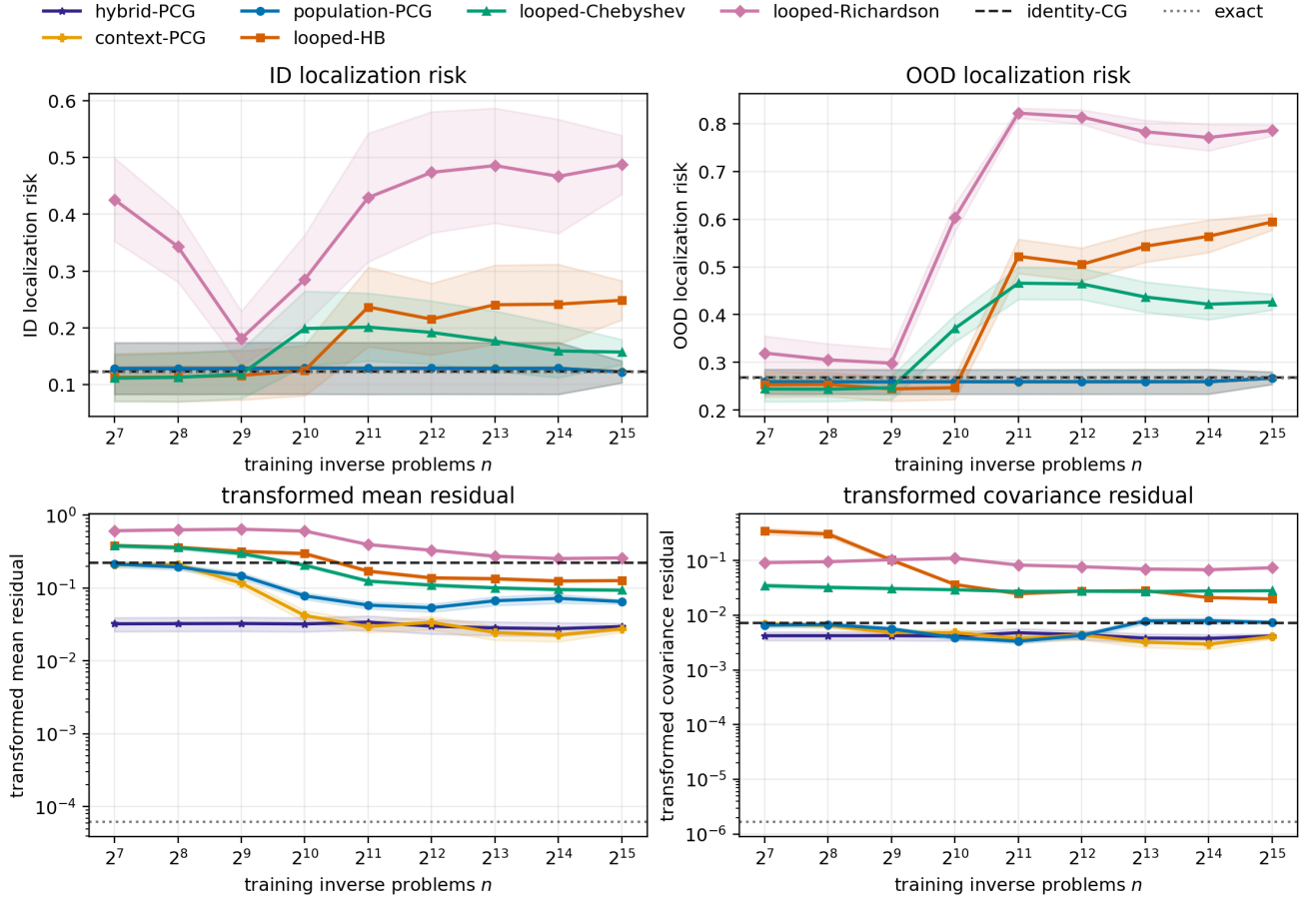


Figure 7: Learning curves in dataset size at fixed network width and context.

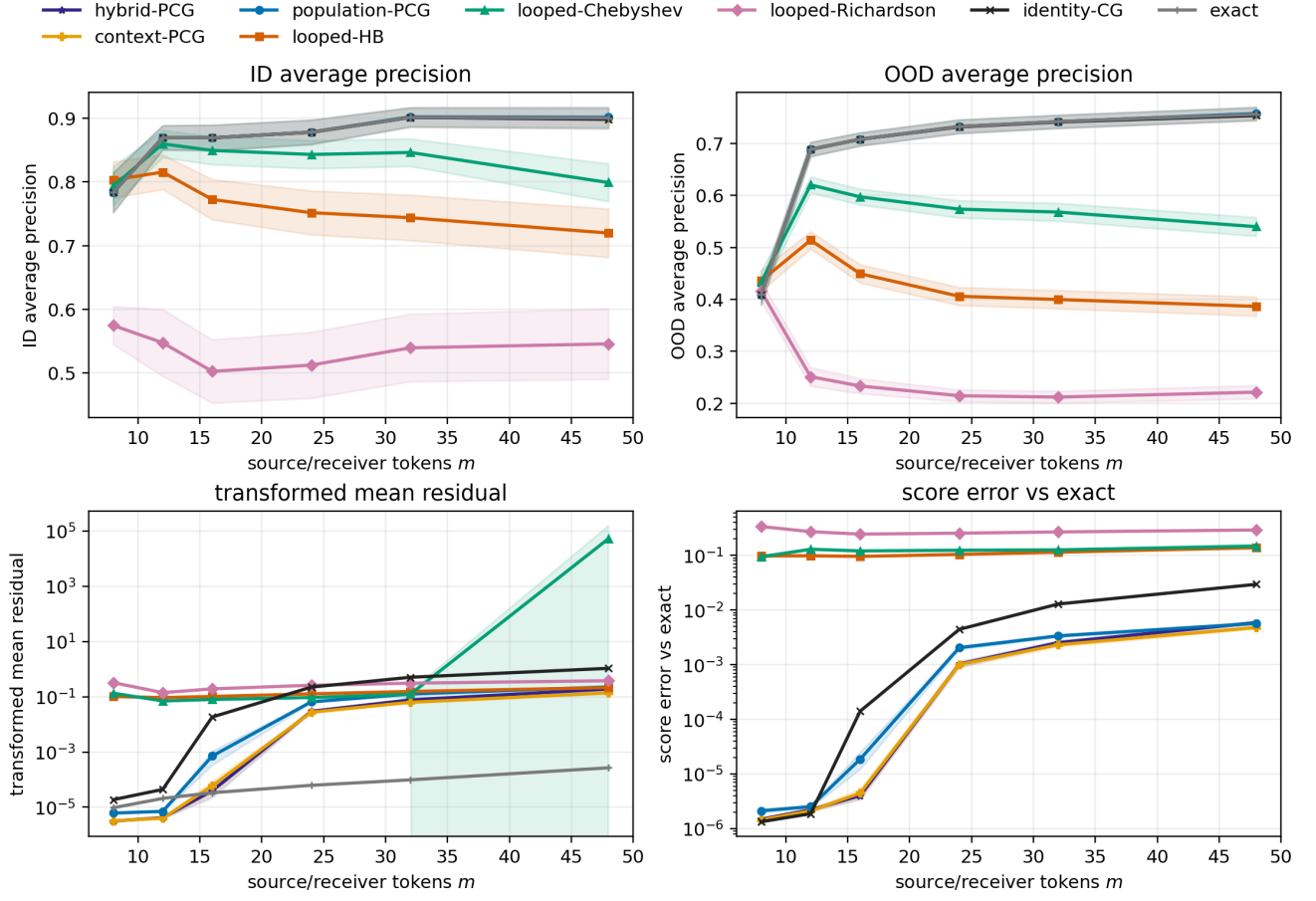


Figure 8: Context-length scaling. Context  $m = 8$  and  $m = 48$  are outside the training context set.

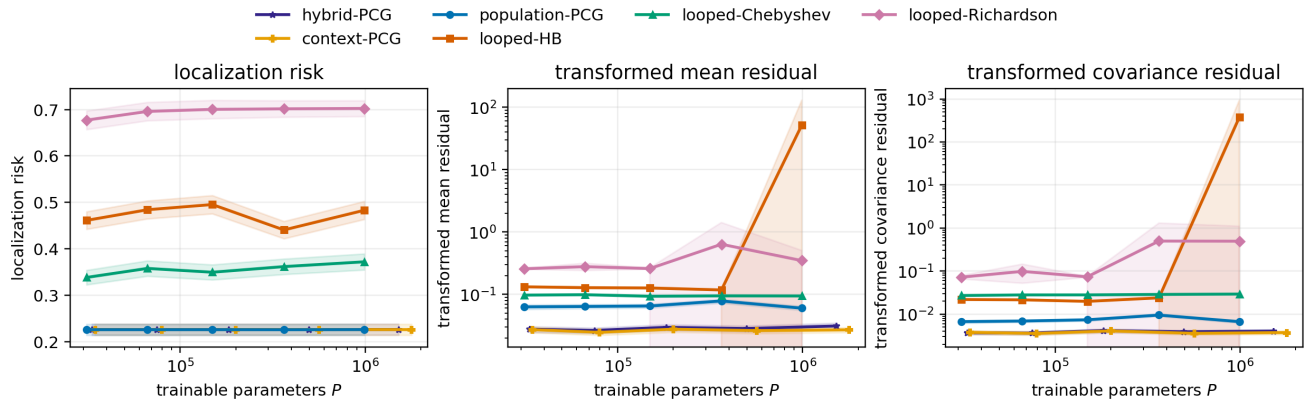


Figure 9: Network-size scaling at the largest completed dataset size.

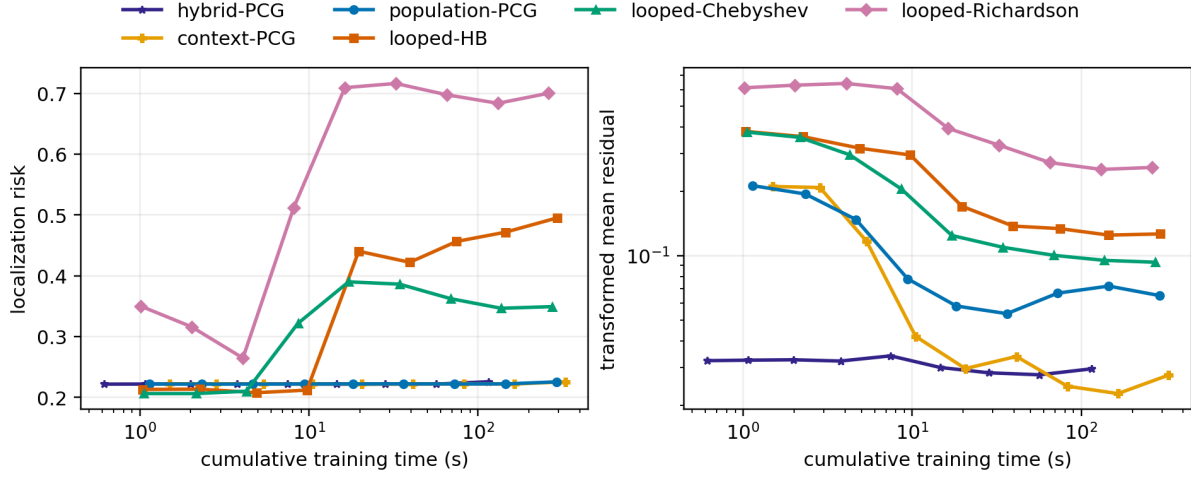


Figure 10: Wall-clock learning curves. Time includes physical task generation, forward/backward propagation and optimization, but excludes held-out audits. Curves use the median over seeds because the accelerator is shared.

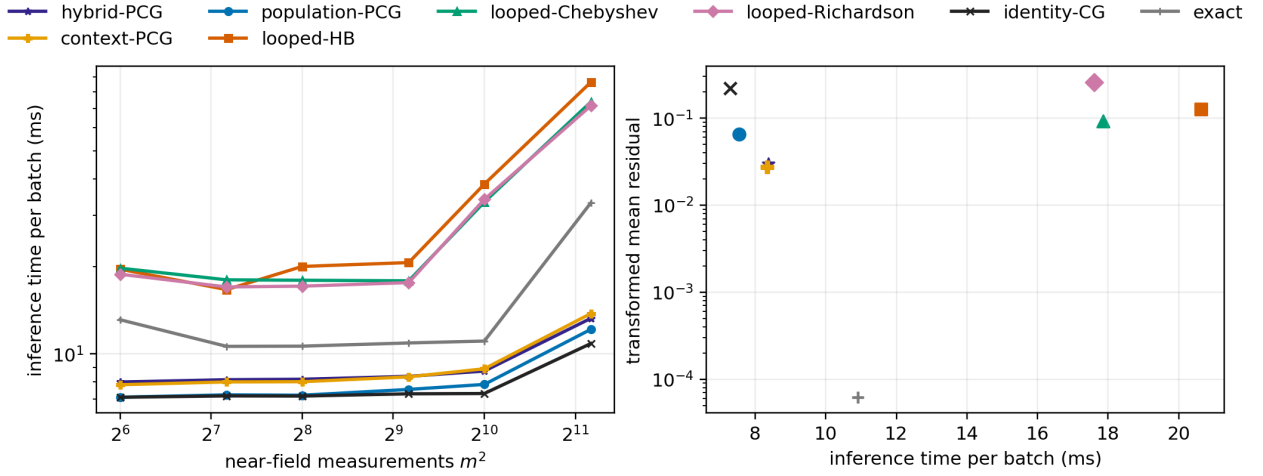


Figure 11: Inference scaling with  $m^2$  complex measurements and the residual-time Pareto comparison at  $m = 24$ .

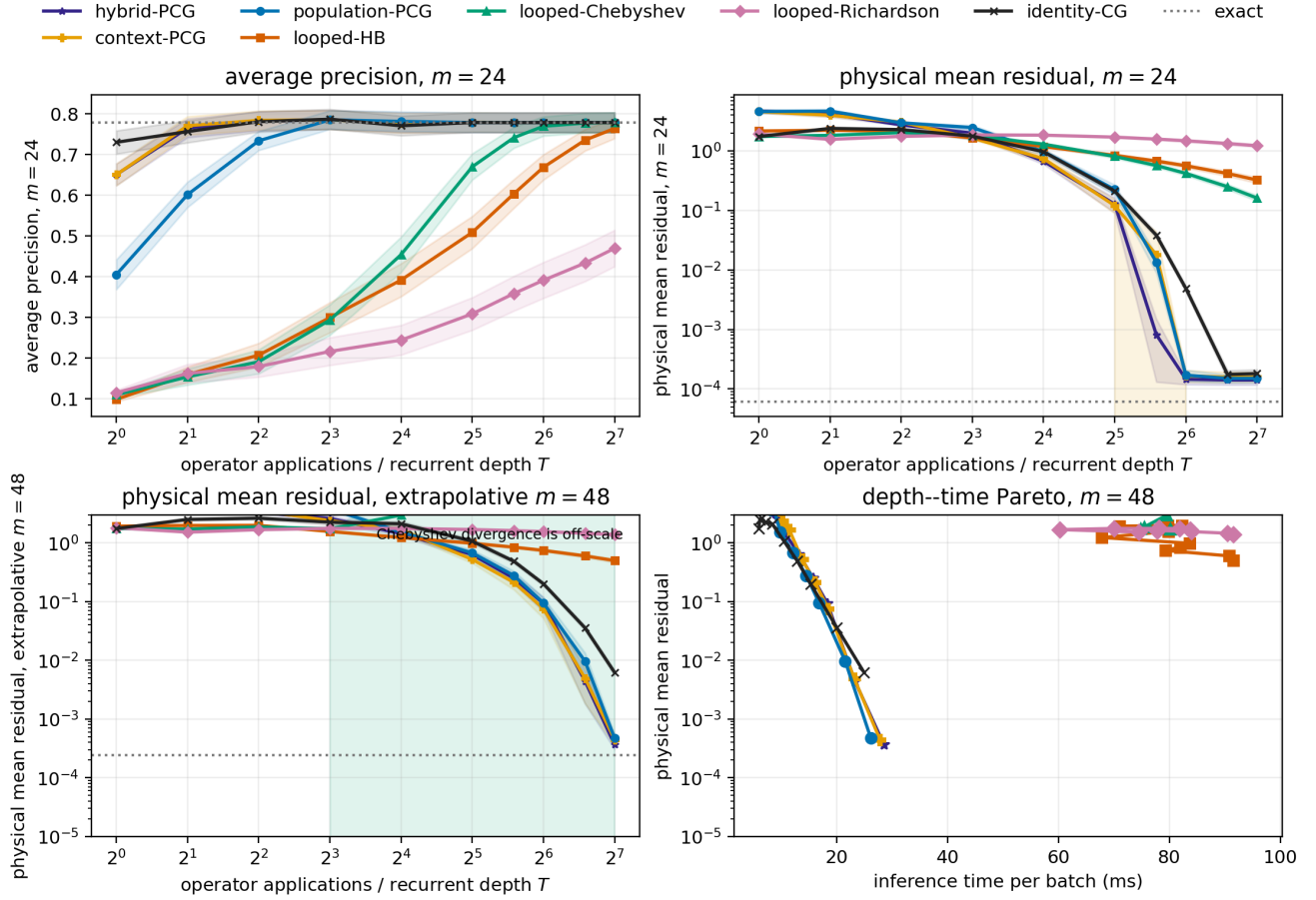


Figure 12: Post-training depth scaling on shared held-out tasks. Varying  $T$  changes only the number of operator applications; parameters and acquisition data are fixed. The lower-right panel reports the realized depth-time Pareto curve rather than assuming ideal linear timing.

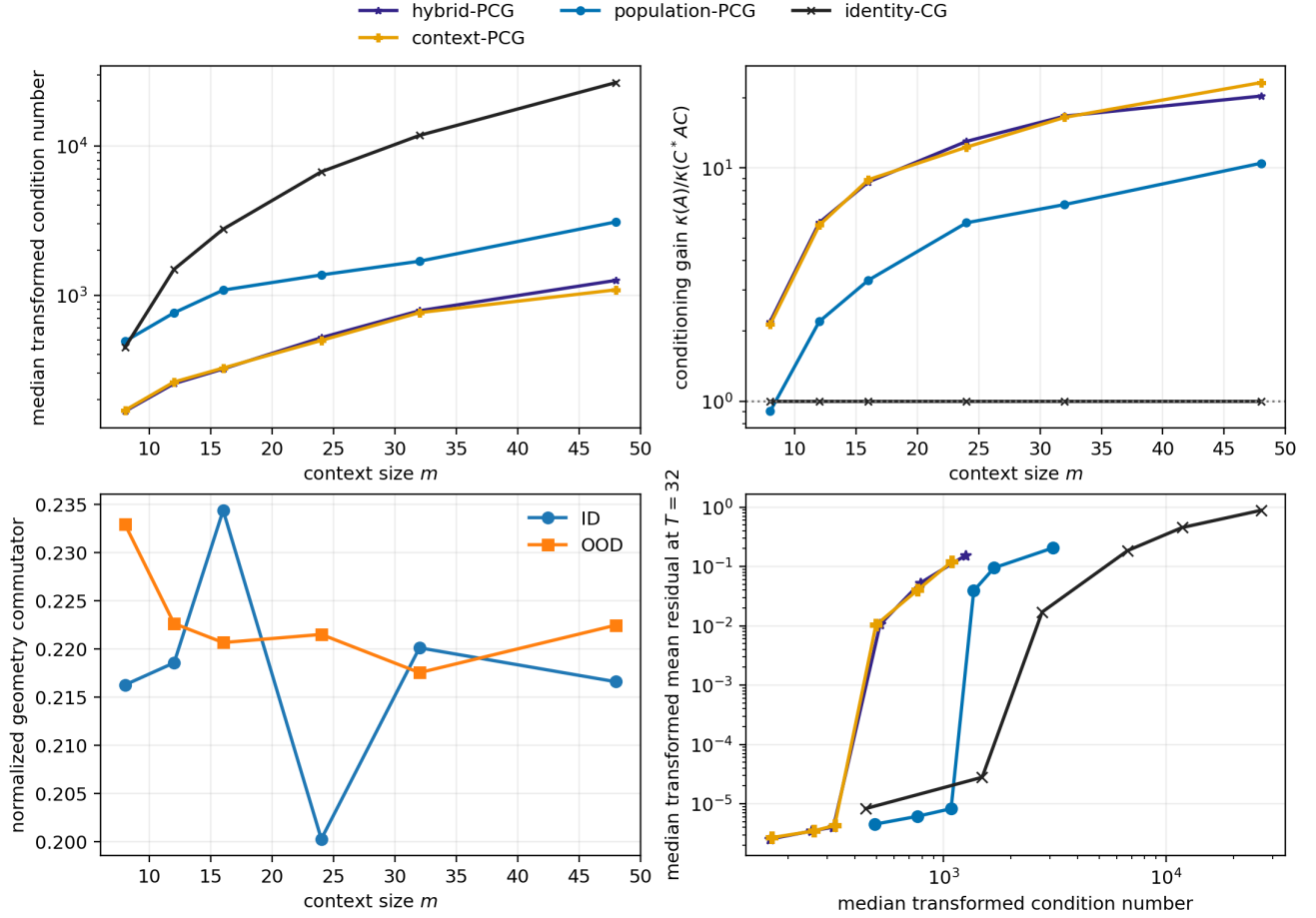


Figure 13: Spectral conditioning audit. The normalized commutator tests the alignment of the fixed angular-kernel basis with the obstacle-dependent near-field Hessian; values away from zero expose a fixed-basis bottleneck.

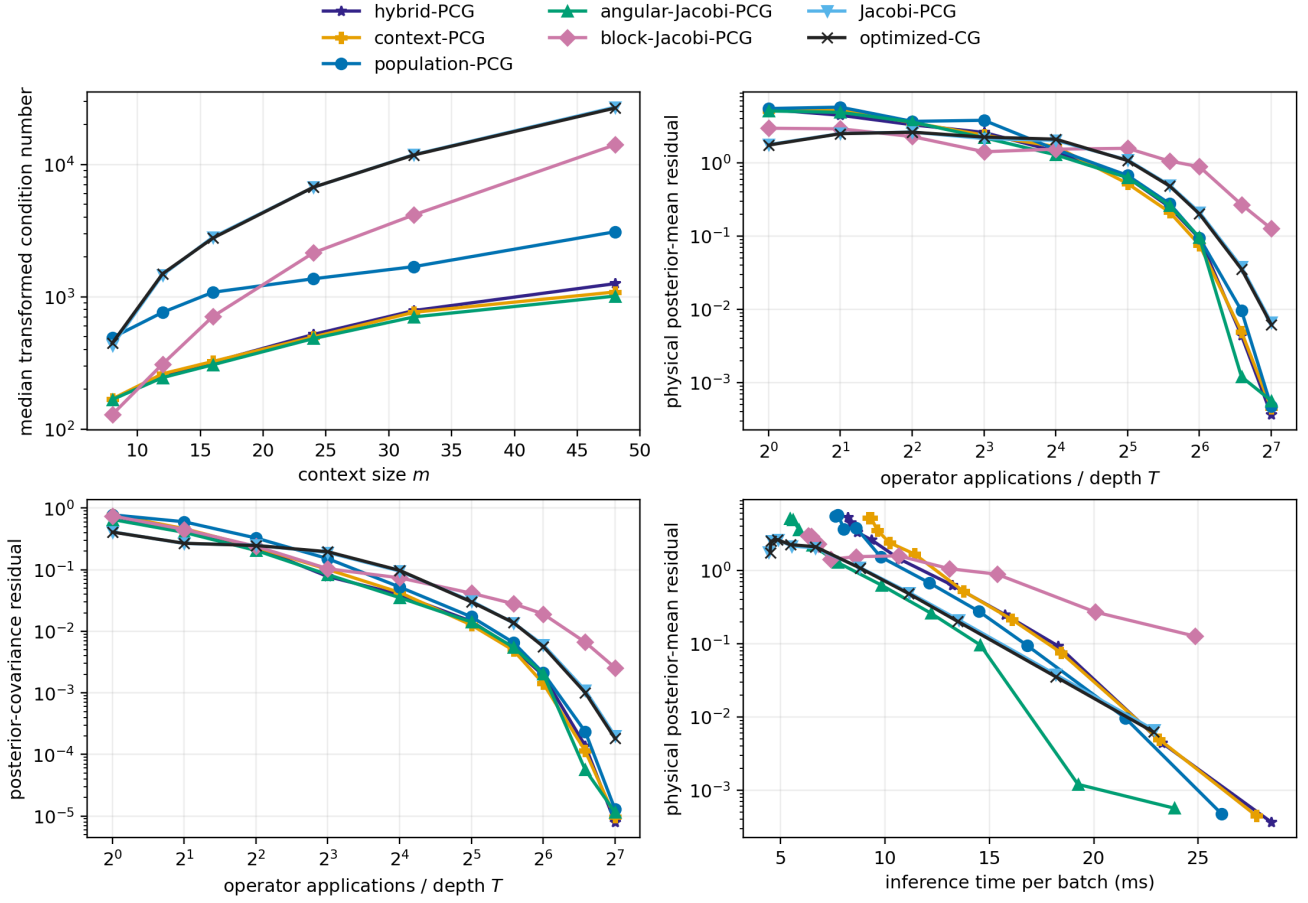


Figure 14: Training-free PCG controls at matched depth. Jacobi uses the physical diagonal, block-Jacobi uses contiguous angular blocks of size four, and angular-Jacobi uses the diagonal of the current Hessian in the fixed GP angular basis.

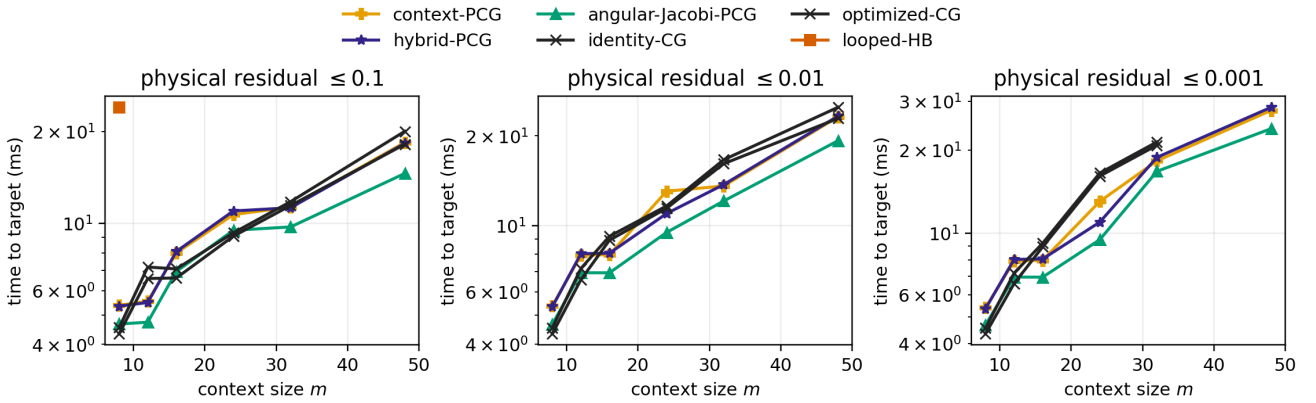


Figure 15: Measured time to residual targets on the audited depth grid. A curve is absent where a method does not reach the target by  $T = 128$ . Equal-depth and time-to-tolerance comparisons need not rank methods identically.

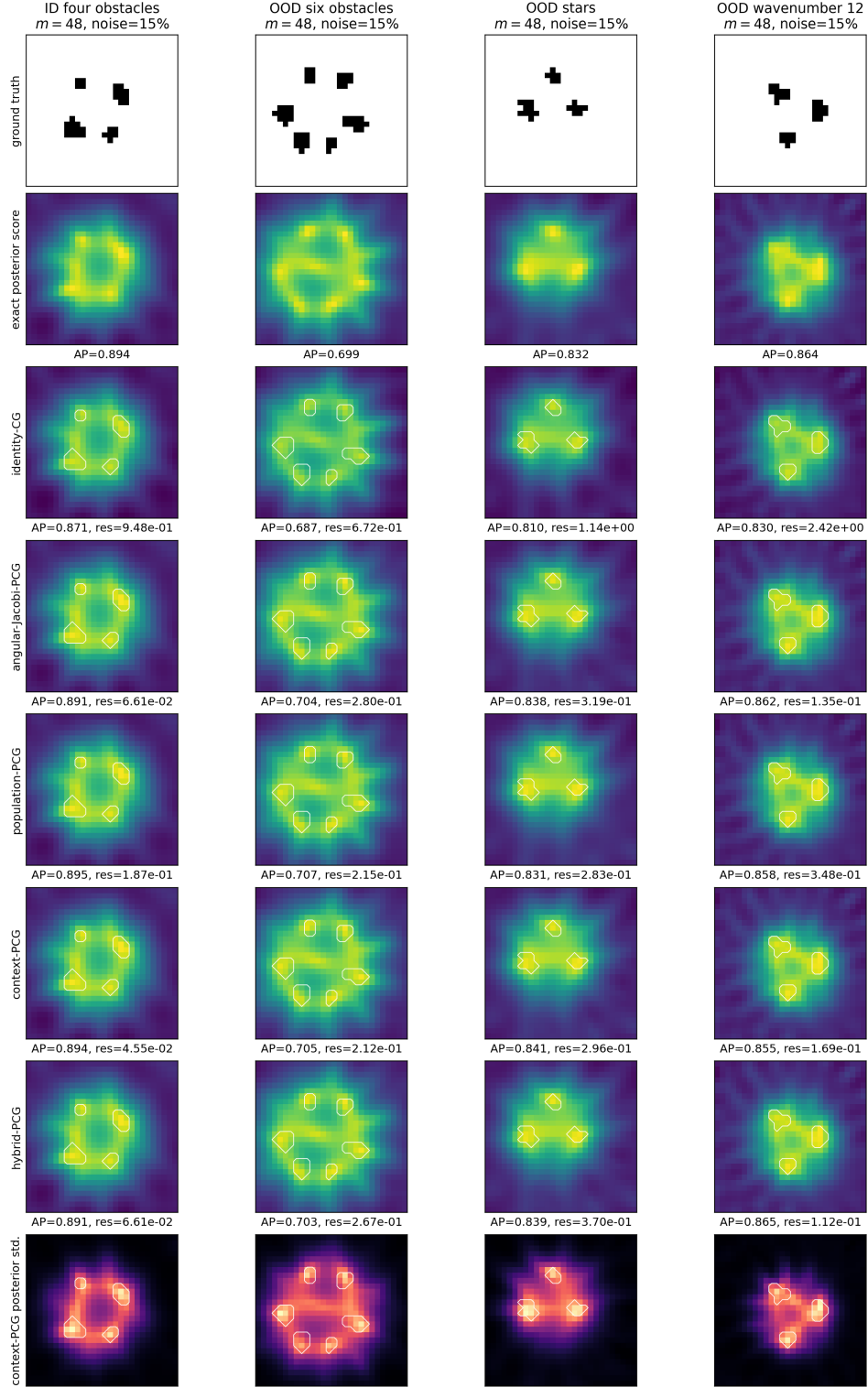


Figure 16: Pre-specified task zero at extrapolative context  $m = 48$ ; no example was selected by performance. White contours show the true obstacles. The rows compare CG, analytic angular-Jacobi PCG, the learned factors, and the analytic-learned hybrid. The last row displays context-PCG posterior standard deviation rather than a reconstruction score.

## 6 Interpretation and limitations

Six claims are deliberately separated. First, CG is a strong task-adaptive baseline because its coefficients already depend on the current Krylov space. Second, population-PCG tests whether geometry-distribution conditioning adds value when no obstacle-specific preconditioner is known. Third, context-PCG tests prompt-dependent modal gains while keeping the angular-kernel basis fixed. Fourth, hybrid-PCG tests whether a bounded learned correction adds to the analytic GP-basis diagonal. Fifth, looped HB tests amortized fixed-depth iteration, not a universally superior replacement for CG. Sixth, angular-Jacobi-PCG tests whether the same task signal can be extracted analytically without training. A fully equivariant task-specific preconditioner would map current moment or Ritz information into an SPD operator  $C_D(A_D) \approx A_D^{-1/2}$ . Neither modal model can rotate all obstacle-dependent eigendirections; a block-Lanczos or polynomial-in- $A_D$  extension is required when the commutator audit is large. The joint-shift audit also shows that nearly equal condition numbers can have different finite-depth residuals. A next loop should therefore encode the right-hand-side-weighted spectral measure through moments  $B^* A_D^j B$ , not only extremal eigenvalues or scalar condition number; exact outer Krylov coefficients can then preserve the CG guarantee.

The held-out bound certifies the stated simulator distribution only. The forward data are generated by MFS, not laboratory measurements, and context extrapolation does not imply robustness to arbitrary receiver geometries. Physical resolution remains controlled by aperture, frequency and sensor count. These qualifications prevent a small numerical loss from being misreported as solved inverse scattering.

## 7 Reproducibility

All task-level records, cumulative training times, checkpoints, runtime measurements, scenario definitions, fitted parameters and bound values are stored beside this PDF. The experiment uses three fixed seeds and preserves the exact same held-out tasks across methods, widths and dataset checkpoints.

## References

- [1] B. Bordelon, M. I. Letey, and C. Pehlevan. Theory of scaling laws for in-context regression: depth, width, context and time. *arXiv:2510.01098*, 2025.
- [2] G. Kang, C. J. Lee, and X. Cheng. Transformers can learn posterior predictive distributions in-context. *arXiv:2605.26713*, 2026.
- [3] J. Garnier, H. Haddar, and H. Montanelli. The linear sampling method for random sources. *arXiv:2210.15560*, 2022.
- [4] J. Garnier, H. Haddar, and H. Montanelli. The linear sampling method for data generated by small random scatterers. *arXiv:2403.19482*, 2024.
- [5] B. T. Polyak. Some methods of speeding up the convergence of iteration methods. *USSR Computational Mathematics and Mathematical Physics*, 1964.
- [6] Y. Saad. *Iterative Methods for Sparse Linear Systems*, second edition, SIAM, 2003.